

SOFTWARE REQUIREMENT SPECIFICATION(SRS)

Project Name: Add Two Number Web App

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Date: 2026 May 9

1.1 INTRODUCTION

The Software Requirement Specification document defines the software requirements for developing the Add Two Numbers web application.

1.2 PURPOSE

The purpose of this system is to provide a simple calculator application that performs addition and stores results in a MySQL database.

1.3 SYSTEM OVERVIEW

The application consists of:

- Frontend
- Backend API
- Database

Architecture Flow:

User → Frontend → FastAPI Backend → MySQL Database

1.4 TECHNOLOGY STACK

Layer	Technology
Frontend	HTML, CSS, JavaScript
Backend	FastAPI
Database	MySQL
ORM	SQLAlchemy
API Testing	Postman
Automation Testing	Selenium

1.5 FUNCTIONAL REQUIREMENTS

Requirement ID	Description
SRS-FR-001	System should accept two numeric inputs
SRS-FR-002	System should provide Add button
SRS-FR-003	System should calculate addition result
SRS-FR-004	System should display result
SRS-FR-005	System should store result in database
SRS-FR-006	System should retrieve calculation history

1.6 NON-FUNCTIONAL REQUIREMENTS

Performance Requirements

- Result should display within 2 seconds.
- API response time should be fast.

Security Requirements

- Input validation should be implemented.
- SQL injection should be prevented.

Usability Requirements

- Interface should be simple.
- User should easily understand workflow.

Reliability Requirements

- System should work continuously without crash.

Compatibility Requirements

- Application should run on modern browsers.

1.7 DATABASE REQUIREMENTS

Database Name

add_app_db

Table Name

calculations

Table Structure

Column Name	Data Type
id	Integer
number1	Float
number2	Float
result	Float

1.8 API REQUIREMENTS

Endpoint: Add Numbers

Method

GET

URL

/add?a=5&b=7

Response

```
{  
  "result": 12  
}
```

Endpoint: History

Method

GET

URL

/history

Response

```
[  
  {  
    "id": 1,  
    "number1": 5,  
    "number2": 7,  
    "result": 12  
  }  
]
```

1.9 TEST REQUIREMENTS

Unit Testing

- Test addition logic.
- Test API responses.

Integration Testing

- Test frontend-backend connection.
- Test backend-database connection.

System Testing

- Test complete application workflow.

Automation Testing

- Automate UI testing using Selenium.

- Automate API testing using pytest.
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1.10 FUTURE ENHANCEMENTS

- Add subtraction feature.
 - Add multiplication feature.
 - Add division feature.
 - Add user login.
 - Add delete history functionality.
 - Deploy application online.
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CONCLUSION

The Add Two Numbers web application is a beginner-level full-stack project that demonstrates frontend development, backend API integration, MySQL database connectivity, and testing concepts. The project can be used for learning software development, manual testing, automation testing, API testing, and database testing.